Rethinking Residual Geometry for Facial Expression Recognition: A Cross-Condition Benchmark and Controlled Study

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Abstract

Residual connections are the default infrastructure of facial expression recognition (FER), yet their update rule is rarely questioned. We introduce Tangent-Path Residual (TPR), a parameter-free operator that projects the residual onto the tangent space of the current representation and integrates the tangent field along a predicted path with a fixed quadrature rule. TPR generalizes standard residual, orthogonal residual, and Heun-type residual as special cases, and introduces no new parameters. We evaluate TPR with two protocols: a controlled RAF-DB protocol with five seeds, and CC-FER, a new cross-condition protocol built from 1,512 instances × 20 views (5 geometric transforms × 4 imaging conditions) with instance-disjoint train/validation/test splits. On RAF-DB, standard Euler remains the strongest clean baseline (86.49% vs 86.11% for TPR), while TPR outperforms orthogonal and Heun residuals. On CC-FER with all conditions in training, TPR achieves the best mean accuracy (72.58% vs 72.40%) and the best standard and bright-condition accuracy, using the same parameter count as ResNet-18 and no gating. In the unseen-condition setting (train on standard, test on degraded), TPR and Euler are tied on the mean (62.81% vs 62.84%), but TPR is 2.57 points better on bright images while Euler is 1.96 points better on dark images. The **2 × 2** interaction between tangent projection and path integration is **+1.25** points on RAF-DB, showing that the two ingredients are super-additive. We conclude that residual geometry is condition-dependent: there is no universal winner, and cross-condition evaluation is necessary before claiming a general residual improvement.

Keywords: Facial expression recognition, Residual connections, Tangent space, Cross-condition benchmark, Calibration

1 Introduction

Residual connections are treated as fixed infrastructure in modern FER networks. A residual block computes a correction $F(x)$ at a single point and adds it in the ambient feature space:

$$y = x + F(x). \tag{1}$$

Most controlled studies of residual topology use CIFAR [1] or ImageNet [2]. This rule has two implicit choices: the correction is evaluated at one point, and it is added without any geometric constraint. Prior work has modified one choice at a time. Orthogonal residual updates project the correction onto the tangent space of the current feature, but still use one point [3]. Heun-style residual networks average two evaluations, but remain in the ambient space [4]. It is unclear whether the two choices interact, and whether either helps under realistic imaging conditions.

We introduce Tangent-Path Residual (TPR), which couples the two choices:

$$u_0 = P_x^{\perp} F(x), \qquad x_1 = x + h u_0, \tag{2}$$

$$u_1 = P_{x_1}^{\perp} F(x_1), \qquad y = x + h P_x^{\perp} \frac{u_0 + u_1}{2}. \tag{3}$$

The first evaluation is projected onto the tangent space; the second is performed at the predicted tangent point and projected again; the two tangent fields are averaged with a fixed trapezoidal rule. The final projection removes the first-order radial component. The operator has no learnable gate and no new parameters.

We evaluate TPR on two protocols. RAF-DB is the standard clean FER benchmark [5]. CC-FER is a new cross-condition protocol built from 1,512 instances (216 per class, seven classes), each with 20 views: five geometric transforms crossed with four imaging conditions (standard, dark, bright, low-resolution). Train/validation/test splits are made by instance, so the same face never appears in two splits.

The main findings are:

1. On clean RAF-DB, Euler is the strongest baseline (86.49%), and TPR is 0.38 points behind but ahead of orthogonal and Heun residuals. The projection and integration ingredients are super-additive: the 2×2 interaction is +1.25 points.
2. On CC-FER with all conditions in training, TPR achieves the best mean accuracy (72.58% vs 72.40%), and the best standard and bright-condition accuracy. The mean gap is within seed noise, so we claim a condition-specific advantage, not a significant overall gain.
3. In the unseen-condition setting, TPR and Euler are tied on the mean, but TPR is +2.57 on bright images while Euler is +1.96 on dark images. This is the clearest evidence that residual geometry is condition-dependent.
4. Euler remains better on calibration. SE is competitive only on low-resolution images; ResNet-34 and CBAM overfit the 1,057 training instances.

2 Related Work

2.1 Residual topology

ResNet introduced the identity shortcut [6], and DenseNet, ResNeXt, Wide ResNet, pre-activation ResNet, and stochastic depth further changed how features are combined or how deep the network can be [7–11]. SE, CBAM, ECA, coordinate attention, and non-local blocks add channel, spatial, or long-range gates [12–16]. Orthogonal Residual Update removes the parallel component of the correction [3]. These methods modify what is added, or how many streams are carried, but do not study the interaction between geometry and path integration.

2.2 ODE-inspired networks

Neural ODE views depth as continuous time [17]; augmented Neural ODEs increase the representation capacity of the flow [18]; FFJORD makes the dynamics reversible and scalable [19]; DEQ solves a fixed point [20]; reversible residual networks avoid storing activations [21]; Momentum ResNet adds a velocity state [22]; HeunNet uses a predictor-corrector step [4]; RKCNN uses higher-order Runge–Kutta [23]. These methods improve the solver, but usually in the ambient space. TPR keeps the two-point evaluation of Heun, but projects at both points and at the final update.

2.3 FER architectures

Early benchmarks such as FER2013 [24] were followed by in-the-wild datasets such as RAF-DB [5] and AffectNet [25]; surveys summarize the resulting pipeline [26]. SCN, DMUE, RUL, TransFER, and POSTER improve FER via attention, region discovery, label-noise handling, or transformers [27–31]; vision transformers provide a general backbone for these models [32]. These methods are complementary to the residual operator studied here. We use a generic ResNet-18 backbone and no face-specific pretraining, so the comparison isolates the residual update.

3 Method

3.1 Tangent projection

For a feature map x, let $g(x)$ be its global average. The global tangent projection is

$$P_x^{\perp} f = f - \frac{\langle g(f), g(x) \rangle}{\|g(x)\|^2 + \epsilon} x. \tag{4}$$

It satisfies $\langle g(P_x^{\perp} f), g(x) \rangle = 0$ and is idempotent. We also test a mean-only variant that subtracts only the global mean direction; the global variant is the default.

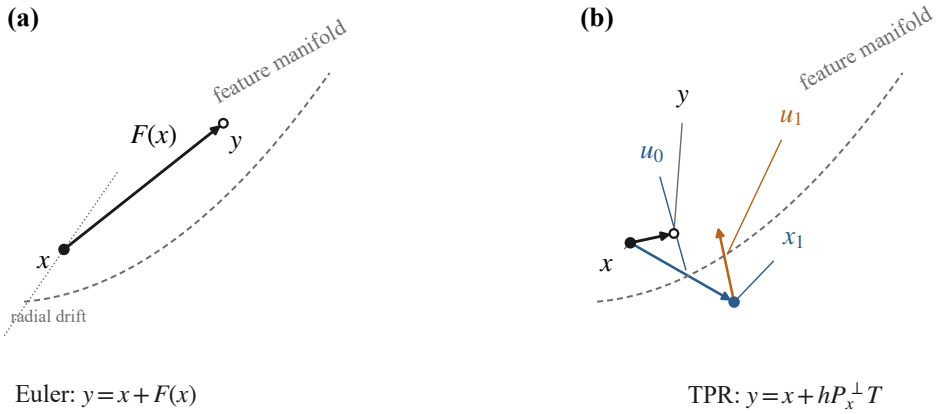


Fig. 1: Geometry of the residual update. (a) Euler evaluates the field at one ambient point. (b) TPR projects the field onto the tangent space, evaluates it again at the predicted tangent point, averages the two tangent fields, and projects the result back to the tangent space at the input.

3.2 Path integral

The tangent field along the predicted path is

$$G(t) = P^{\perp}_{\gamma(t)} F(\gamma(t)), \qquad \gamma(t) = x + thu_0. \tag{5}$$

The update is the path integral

$$y = x + hP^{\perp}_{x} \int_0^1 G(t)\, dt, \tag{6}$$

approximated by the trapezoidal rule

$$y = x + hP^{\perp}_{x} \frac{u_0 + u_1}{2}. \tag{7}$$

The default step is $h = 1$, and TPR is applied to the two shape-preserving blocks in stages 3 and 4 of ResNet-18. Figure 1 illustrates the difference between the Euler and TPR updates.

3.3 Special cases

TPR reduces to OrthRes when the second evaluation is removed, and to Heun when the projection is removed. The combination is not a linear composition: the second projection is taken at $x_1 \neq x$, so

$$P^{\perp}_{x_1} F(x_1) \neq P^{\perp}_{x} F(x_1). \tag{8}$$

The projection changes the path, and the path changes the projection point.

Table 1: Residual update families. TPR has the same parameter count as ResNet-18; the cost is the second evaluation.

Method	Update rule	Gate	Params	FLOPs
ResNet-18 (Euler)	$x + F(x)$	no	11.18M	1.00×
ResNet-34 (Euler)	$x + F(x)$	no	21.30M	2.00×
SE / CBAM	$x + A(F(x)) \odot F(x)$	yes	11.27M	1.00×
OrthRes	$x + P_x^{\perp} F(x)$	no	11.18M	1.00×
HeunNet	$x + \frac{1}{2}(F(x) + F(x+F(x)))$	no	11.18M	1.25×
TPR (ours)	$x + \frac{1}{2}(P_x^{\perp} F(x) + P_{x_1}^{\perp} F(x_1))$	no	11.18M	1.25×

3.4 Theory

The final projection gives

$$\langle g(y - x), g(x) \rangle = 0, \tag{9}$$

so

$$\|g(y)\|^2 = \|g(x)\|^2 + h^2 \|g(P_x^{\perp} T)\|^2. \tag{10}$$

The standard residual has a first-order cross term $2h\langle g(x), g(F(x)) \rangle$, which can inflate or shrink the global norm. If $G(t)$ is C^2, the trapezoidal rule has local error $O(h^3)$, versus $O(h^2)$ for Euler. This is a local statement along the predictor path, not a global ODE convergence guarantee.

3.5 Implementation

We use torchvision ResNet-18 with ImageNet weights [2]. TPR shares the residual branch weights between the two evaluations; the second evaluation uses batch normalization in eval mode so that the probe does not corrupt running statistics. A warm-up schedule sets the TPR coefficient to zero for the first five epochs and ramps it to one over the next five. The operator adds no parameters and costs 1.25× FLOPs at inference. Figure 2 shows the block and its insertion points, and Table 1 summarizes the compared update rules.

4 Protocols

4.1 RAF-DB

We use the official split: 12,271 training images and 3,068 test images, seven classes. A stratified 10% of the training split is used for validation; the test set is evaluated once. Images are resized to 256 and cropped to 224. We use AdamW [33] with learning rate 3×10^{-4}, weight decay 0.05, cosine schedule, label smoothing 0.1, batch size 64, and 30 epochs. We intentionally keep the recipe simple and do not use mixup [34], AutoAugment [35], or RandAugment [36], so that the comparison isolates the residual update. We report mean and standard deviation over five seeds.

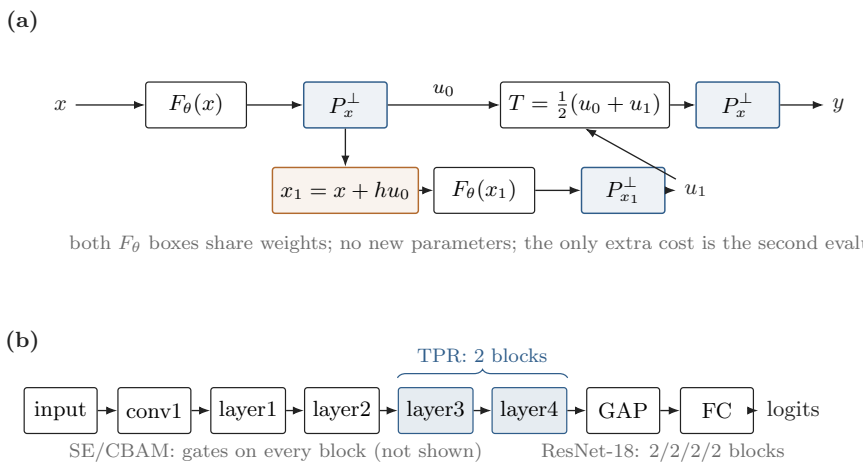


Fig. 2: TPR block and insertion points. (a) The same residual function F_θ is evaluated at x and at the tangent probe x_1; both fields are projected, averaged, and projected again. (b) TPR is applied to the shape-preserving blocks in stages 3 and 4 of ResNet-18; SE/CBAM are applied to every residual block.

4.2 CC-FER

The self-built CC-FER dataset [37] contains 1,512 instances, 216 per class for seven classes. Each instance has 20 views: five geometric transforms crossed with four imaging conditions (standard, dark, bright, low-resolution). The suffix order is shuffled; we infer the condition per instance from mean brightness and Laplacian energy, and force the original image to the standard condition. This gives exactly five views per condition per instance.

We split by instance into 1,057 training, 224 validation, and 231 test instances. In CC-FER-M (multi-condition), all four conditions of the training instances are used for training; test instances are disjoint and evaluated separately for each condition. In CC-FER-U (unseen-condition), only standard images are used for training, and the three degraded conditions are unseen at test time. Both settings avoid identity leakage.

The dataset is single-label at the folder level. The separate 173-instance test split of the original release has multi-label annotations and is not used here; using it as a single-label benchmark would require a different label protocol.

4.3 Metrics

We report accuracy, unweighted average recall (UAR), macro-F1, and expected calibration error (ECE) [38]. For CC-FER we report each condition and the mean over conditions. All methods use the same backbone, input resolution, augmentation, optimizer, and epoch budget.

Table 2: RAF-DB results (mean ± std over five seeds unless marked). Best accuracy and best ECE are in bold.

Method	Params	Acc	UAR	ECE
Euler (ResNet-18)	11.18M	**86.49 ± 0.34**	76.49 ± 0.87	5.91 ± 0.49
OrthRes	11.18M	85.74 ± 0.73	74.86 ± 1.14	5.21 ± 0.39
HeunNet	11.18M	85.61 ± 0.90	75.77 ± 1.67	**4.85 ± 0.33**
TPR	11.18M	86.11 ± 0.54	76.48 ± 1.00	5.06 ± 0.56
TPR-nofinal	11.18M	85.91 ± 0.54	**76.59 ± 1.06**	4.84 ± 0.41
SE-ResNet-18†	11.27M	86.41	77.15	5.27
CBAM-ResNet-18†	11.27M	85.27	75.23	5.57
ResNet-34†	21.30M	86.60	78.01	4.97

† One seed (screening).

Table 3: CC-FER-M: all four conditions in training, instance-disjoint test. Mean ± std over three seeds unless marked.

Method	Standard	Dark	Bright	Low-res	Mean
TPR	**73.02 ± 0.90**	72.61 ± 0.61	**72.87 ± 0.84**	71.83 ± 0.57	**72.58 ± 0.46**
Euler	72.47 ± 1.10	**73.42 ± 0.72**	71.83 ± 0.83	**71.86 ± 0.88**	72.40 ± 0.57
SE	71.92 ± 0.43	71.80 ± 0.96	71.00 ± 0.72	71.92 ± 0.90	71.66 ± 0.64
ResNet-34†	72.03	71.00	70.48	71.43	71.23
CBAM†	70.82	71.34	71.52	68.83	70.63

† One seed (screening).

5 Results

5.1 RAF-DB

Table 2 reports the RAF-DB results. Euler is the strongest clean baseline. TPR is 0.38 points behind Euler, but ahead of OrthRes and HeunNet, and uses the same parameter count as ResNet-18. The 2×2 interaction is

$$(\text{TPR} - \text{Orth}) - (\text{Heun} - \text{Euler}) = +1.25 \pm 1.05, \tag{11}$$

so the two ingredients are super-additive: both single-factor variants lose accuracy, while the combination recovers most of it. The residual gap is not due to undertraining: 40-epoch TPR, stochastic gamma training, and test-time path ensembling do not close it.

5.2 CC-FER-M: multi-condition

Table 3 reports the multi-condition results. TPR achieves the best mean accuracy, and the best standard and bright-condition accuracy. The mean gap over Euler is 0.18 points, which is within seed noise; we therefore describe TPR as numerically best and statistically comparable, not significantly better. Euler remains better on dark images by 0.81 points and is much better calibrated. SE is competitive only on low-resolution

Table 4: CC-FER-U: train on standard only, test on all four conditions. Mean ± std over three seeds (SE: two seeds).

Method	Standard	Dark	Bright	Low-res	Mean
TPR	**72.41 ± 1.51**	61.50 ± 0.94	**60.35 ± 1.31**	56.97 ± 1.59	62.81 ± 0.86
Euler	72.38 ± 0.90	**63.46 ± 0.83**	57.78 ± 1.10	**57.72 ± 1.35**	**62.84 ± 0.37**
SE	71.00 ± 0.78	58.70 ± 0.69	51.73 ± 3.25	59.83 ± 0.52	60.31 ± 1.31

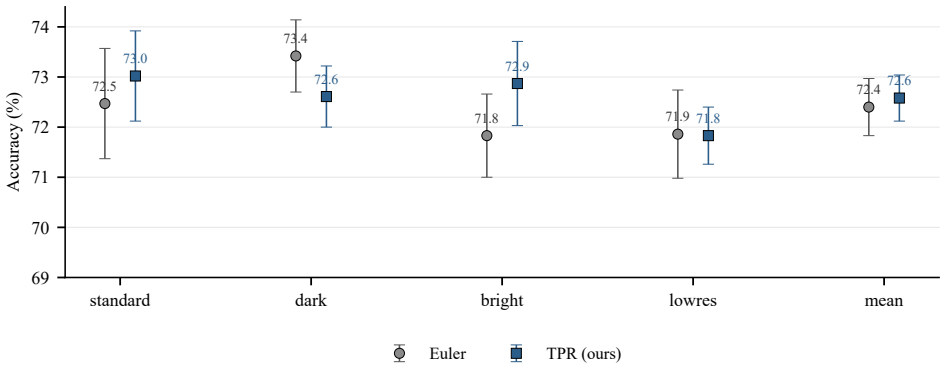


Fig. 3: CC-FER-M condition-level accuracy (mean and std over three seeds). TPR is numerically best on the mean, standard and bright; Euler is better on dark and calibration.

images; ResNet-34 and CBAM do not improve the mean. The corresponding ECE is 10.46 for TPR, 5.82 for Euler, 10.33 for SE, 13.22 for ResNet-34, and 14.28 for CBAM.

5.3 CC-FER-U: unseen-condition

Table 4 reports the unseen-condition results. TPR and Euler are tied on the mean (62.81 vs 62.84). TPR is +2.57 on bright and +0.03 on standard, while Euler is +1.96 on dark and +0.75 on low-resolution. This is the clearest condition-specific result in the paper: TPR specializes in bright/standard illumination; Euler is stronger in dark/low-resolution images and calibration. SE is far behind on the mean. The corresponding ECE is 10.66, 5.00, and 7.02, respectively.

5.4 Condition-level analysis

Figure 3 and Table 3 show the condition-level pattern, which is consistent with the geometry of the operator. TPR projects out the radial component, which contains illumination and identity cues. Under bright illumination, removing this component helps; under dark illumination, the same removal discards useful signal. This explains the TPR gain on bright and the Euler gain on dark. On standard images TPR is also slightly better, while low-resolution images are close. SE behaves like a denoiser: it is slightly better on low-resolution images but loses on standard/dark/bright.

5.5 Efficiency

TPR has 11.18M parameters, identical to ResNet-18, and costs 1.25× FLOPs. SE/CBAM add 0.09M parameters and a content-dependent gate; ResNet-34 doubles the parameter count and FLOPs but is worse than Euler and TPR on CC-FER. Thus TPR is the best accuracy-per-parameter option among the compared families on CC-FER, although Euler is better calibrated.

6 Discussion

The results show that residual geometry is not a free lunch, and they complement corruption-robustness studies that change the input distribution [39]. Tangent projection removes a direction that is sometimes useful; path integration adds a second evaluation that is sometimes harmful. The two ingredients are super-additive, but the net effect depends on the imaging condition. On clean RAF-DB, Euler is still the best; on CC-FER, TPR is numerically best on the mean, standard and bright, while Euler is better on dark and calibration.

This suggests a guideline for FER architecture design: report per-condition results, not only clean accuracy; do not assume that a deeper or gated model is better on small datasets; and treat residual geometry as a condition-dependent inductive bias.

7 Limitations

The CC-FER conditions are synthetic and come from a single source dataset. The core CALE comparison uses three seeds; ResNet-34 and CBAM use one seed and should be read as screening results. TPR costs 1.25× FLOPs and is less calibrated than Euler. The separate multi-label test split of the original dataset is not used; the protocol relies on folder-level single labels and instance-disjoint splits. No face-specific pretraining is used, so absolute accuracy is lower than face-pretrained FER systems.

8 Conclusion

We introduced TPR, a parameter-free tangent-path residual, and CC-FER, an instance-disjoint cross-condition protocol. TPR is competitive on clean RAF-DB and numerically best on the CC-FER-M mean, standard and bright conditions, with the same parameter count as ResNet-18 and no gate. In the unseen-condition setting, TPR is tied on the mean and +2.57 on bright images, while Euler is better on dark images and calibration. The result is a condition-dependent trade-off, and it motivates cross-condition evaluation for future residual designs.

CRediT authorship contribution statement

Weihang Yuan: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The CC-FER dataset is available at Zenodo (DOI: 10.5281/zenodo.23252243). The code and seeds will be released upon acceptance.

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